Validation Criterion Formulation Module (VCFM)

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Governed Space

Validation Criterion Formulation

Canonical Definition

Validation Criterion Formulation Module (VCFM) defines the governance framework for transforming qualified Validation Requirements into explicit, evaluable, bounded, interpretable, and traceable Validation Criteria capable of supporting subsequent validation methodology, evidence assessment, execution, and determination.

It establishes the conditions under which a Validation Requirement becomes a formally governed evaluation condition against which a defined characteristic, property, behavior, state, capability, relationship, or outcome of the Validation Object can later be assessed.

Operational Role

VCFM governs the criterion-formulation layer of Validation Criteria governance.

Its role is to convert:

Qualified Validation Requirement → Governed Validation Criterion

A Validation Requirement identifies a condition that matters.

A Validation Criterion expresses that condition in a form that can actually be evaluated.

VCFM therefore creates the methodological bridge between requirement derivation and formal validity assessment.

Module Operational Space

VCFM governs:

- criterion formulation,
- criterion specification,
- criterion evaluability,
- criterion clarity,
- criterion precision,
- criterion boundaries,
- criterion interpretation,
- criterion relevance,
- criterion applicability,
- criterion granularity,
- criterion independence,
- composite criteria,
- conditional criteria,
- criterion normalization,
- criterion traceability,
- and criterion maintenance.

Module Function

VCFM applies after relevant Validation Requirements have been sufficiently derived and qualified.

Its function is to prevent:

- vague requirements being treated as executable criteria,
- ambiguous evaluation conditions,
- criteria incapable of objective assessment,
- multiple requirements being unintentionally collapsed into one criterion,
- one requirement producing incompatible criteria without justification,
- undefined terminology influencing validation,
- criteria being formulated around available results,
- criterion wording predetermining a preferred outcome,
- and validation conclusions being based upon conditions that were never explicitly evaluable.

Requirement and Criterion Distinction

VCFM preserves a fundamental methodological distinction:

Validation Requirement = what condition must be represented within validation.

Validation Criterion = the explicit evaluation condition through which that requirement can be assessed.

For example:

Requirement: The system must remain sufficiently accurate for its approved operational use.

This requirement alone may not be directly evaluable.

VCFM transforms it into one or more explicit criteria defining the characteristic to be evaluated and the conditions under which satisfaction can subsequently be determined.

Therefore:

Requirement → Formulation → Criterion

not:

Requirement = Criterion

Criterion Formulation Requirements

A governed Validation Criterion should establish, where applicable:

Evaluated Characteristic

The property, behavior, state, capability, relationship, outcome, or other characteristic being evaluated must be identifiable.

Evaluation Condition

The condition against which that characteristic will be assessed must be explicit.

Applicability

The Validation Object, component, use, environment, population, configuration, or other condition to which the criterion applies must be identifiable.

Evaluation Boundary

The criterion must be sufficiently bounded to distinguish what it evaluates from what it does not evaluate.

Interpretability

The meaning of the criterion must be sufficiently clear for subsequent methodology and determination.

Traceability

The criterion must remain connected to the qualified Validation Requirement from which it was derived.

Criterion Evaluability

A Validation Criterion must be capable of meaningful evaluation.

Evaluability does not require every criterion to be purely numerical.

A criterion may be evaluated through:

- quantitative measurement,
- qualitative assessment,
- categorical determination,
- documentary verification,
- experimental observation,
- statistical analysis,
- expert assessment,
- formal verification,
- simulation,
- or another legitimate validation method.

VCFM does not select that method.

It ensures that the criterion is formulated clearly enough for a suitable method to be selected later.

Criterion Precision

Criterion precision should be proportionate to the Validation Purpose and required Validation Depth.

Excessively broad criteria may create interpretation ambiguity.

Excessively narrow criteria may fail to represent the underlying Validation Requirement.

VCFM therefore seeks sufficient precision without unnecessarily prescribing implementation.

Criterion Granularity

A Validation Requirement may produce:

- one criterion,
- multiple independent criteria,
- multiple dependent criteria,
- a composite criterion,
- or a hierarchical set of criteria.

Criterion granularity should reflect the structure of the underlying requirement.

Where materially different conditions can independently affect validation, separating them may improve traceability and determination quality.

Atomic Criteria

Where appropriate, criteria should be sufficiently atomic that their satisfaction can be meaningfully evaluated without hiding multiple unrelated conditions inside one evaluation statement.

For example, a criterion requiring a system to be simultaneously:

accurate, secure, reliable, fair, and explainable

may represent several distinct validation conditions rather than one coherent criterion.

VCFM encourages separation where doing so materially improves evaluability and governance clarity.

Composite Criteria

Some Validation Requirements legitimately depend upon multiple subordinate conditions.

VCFM allows creation of **Composite Validation Criteria** where:

- subordinate criteria are identifiable,
- their relationships are defined,
- aggregation logic is explicit,
- and the composite condition remains methodologically interpretable.

A composite criterion must not conceal failure or uncertainty within one of its material components.

Conditional Criteria

Some criteria apply only when particular conditions exist.

Conditional activation may depend upon:

- intended use,
- operational environment,
- system configuration,
- population,
- jurisdiction,
- risk level,

- dependency state,
- lifecycle stage,
- or another governed condition.

VCFM requires activation conditions to remain explicit.

A conditional criterion must not silently appear or disappear during validation.

Criterion Independence

Where criteria are intended to represent independent validation conditions, VCFM should preserve sufficient separation between them.

Duplicative or heavily overlapping criteria may distort the apparent strength or completeness of validation.

For example, the same underlying condition should not be counted repeatedly under differently worded criteria merely to create an impression of stronger validation coverage.

Criterion Neutrality

Validation Criteria should be formulated before the relevant validation results are known wherever reasonably possible.

Criterion wording must not be manipulated to favor:

- validation success,
- validation failure,
- a preferred system,
- a preferred provider,
- a preferred decision,
- or another predetermined outcome.

VCFM therefore protects the separation between:

evaluation conditions and **observed results.**

Criterion Normalization

Where multiple criteria use inconsistent terminology, structures, units, classifications, or evaluation expressions, VCFM may support normalization.

Normalization may improve:

- comparability,
- interoperability,
- machine readability,

- cross-system analysis,
- criteria mapping,
- and validation automation.

Normalization must not materially alter the substantive meaning of the originating Validation Requirement.

Minimum Implementation Framework

1. Receive Qualified Validation Requirement

Confirm that the requirement has sufficient relevance, applicability, authority, and traceability to support criterion formulation.

2. Identify the Evaluated Characteristic

Determine what property, state, behavior, capability, relationship, outcome, or other characteristic must be evaluated.

3. Formulate the Evaluation Condition

Express the requirement as an explicit condition capable of subsequent assessment.

4. Establish Criterion Boundary and Applicability

Determine precisely where and under which governed conditions the criterion applies.

5. Assess Evaluability

Determine whether the criterion can meaningfully support selection of a validation method and subsequent determination.

6. Establish Criterion Structure

Determine whether the criterion should remain:

- atomic,
- composite,
- conditional,
- dependent,
- or otherwise structurally classified.

7. Preserve Criterion Traceability

Maintain:

- Criterion Identifier,
- originating Validation Requirement,

- evaluated characteristic,
- criterion statement,
- applicability,
- boundary,
- structural classification,
- interpretation notes where required,
- version,
- changes,
- and sufficient lifecycle traceability.

Criterion Reformulation

A criterion may require reformulation where:

- its meaning is ambiguous,
- it cannot be meaningfully evaluated,
- its boundary is unclear,
- it combines materially unrelated conditions,
- it fails to represent its originating requirement,
- its terminology has become obsolete,
- or later governance reveals a structural deficiency.

Material reformulation must be versioned and its effect upon downstream validation activity assessed.

Criterion Change Governance

Once validation execution has begun, material criterion change becomes a governance event.

Changes must not be silently applied to previously generated results.

Where a criterion changes materially, VALIDOS® should determine whether:

- existing evidence remains applicable,
- existing execution remains sufficient,
- affected results require reassessment,
- the Validation Determination must be reconsidered,
- or revalidation is necessary.

Criterion-to-Method Transition

VCFM ends where Validation Method governance begins.

The output of VCFM should make it possible for the subsequent Parent Standard to answer:

Which validation method can legitimately determine whether this criterion has been satisfied?

The governed progression therefore becomes:

Validation Requirement → Validation Criterion → Validation Method

This separation is essential to VALIDOS®.

The criterion defines **what must be evaluated.**

The method determines **how it will be evaluated.**

Use Case 1 — AI Output Accuracy

Scenario

A Validation Requirement states that an AI system must provide sufficiently accurate outputs for a defined operational use.

Application VCFM determines that "sufficiently accurate" is not yet an adequately formulated Validation Criterion.

The relevant output characteristic, applicable use, evaluation boundary, and criterion structure are explicitly established.

Result

A subsequent validation method can be selected against an explicit evaluation condition rather than an undefined expectation of "accuracy."

Use Case 2 — Autonomous System Safety

Scenario

A requirement states that an autonomous system must operate safely within its approved environment.

Application

VCFM identifies that the requirement contains multiple potentially independent safety conditions.

Where necessary, these are formulated into separate or composite Validation Criteria with explicit applicability and boundaries.

Result

Safety validation becomes a structured set of evaluable conditions rather than a single broad statement whose satisfaction cannot be objectively demonstrated.

Canonical Closing Statement

Validation Criterion Formulation Module establishes the canonical governance framework for transforming qualified Validation Requirements into explicit and evaluable Validation Criteria. By governing criterion specification, clarity, precision, boundaries, applicability, granularity, independence, composite structures, conditional activation, neutrality, normalization, change, and traceability, VCFM ensures that every validation judgment can ultimately be connected to a clearly formulated evaluation condition established before the result it is intended to assess.

Parent Resources

→ VALIDOS® Validation Governance Architecture — Validation Governance Layer

→ VALIDOS® Architecture Map

→ VALIDOS® Complete Standards & Modules Index

Internal Module Flow

→ Previous module: Validation Requirement Derivation Module (VRDM)

→ Next module: Validation Threshold Calibration Module (VTCM)

Related Documents

→ Validation Criteria Standard

→ About Validation Criteria Standard

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